

SAT 9

Math Module 2

Question # **ID**
SAT9 M 2.1 c841e8e8

$$k + 12 = 336$$

What is the solution to the given equation?

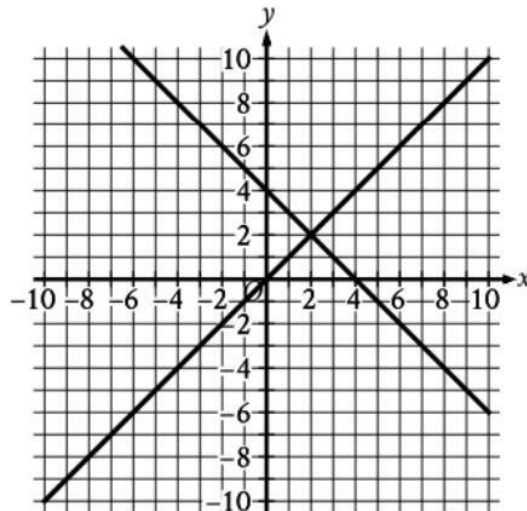
- A. 28
- B. 324
- C. 348
- D. 4,032

SAT9 M 2.2 2cdefcb1

What length, in centimeters, is equivalent to a length of 51 meters? (1 meter = 100 centimeters)

- A. 0.051
- B. 0.51
- C. 5,100
- D. 51,000

SAT9 M 2.3 a09ma203



The graph of a system of two linear equations is shown. What is the solution (x, y) to the system?

(A) (0, 4)

(C) (4, 0)

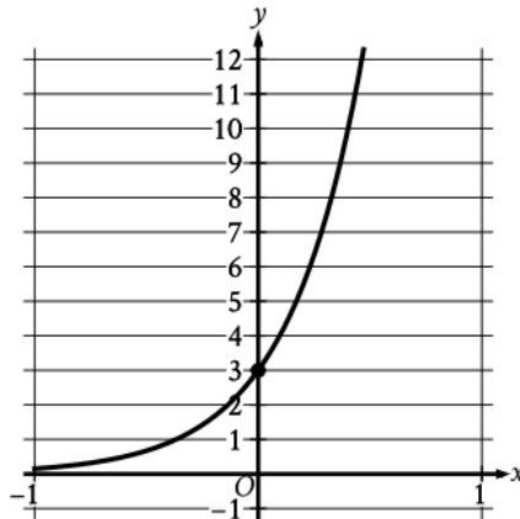
(B) (2, 2)

(D) (4, 4)

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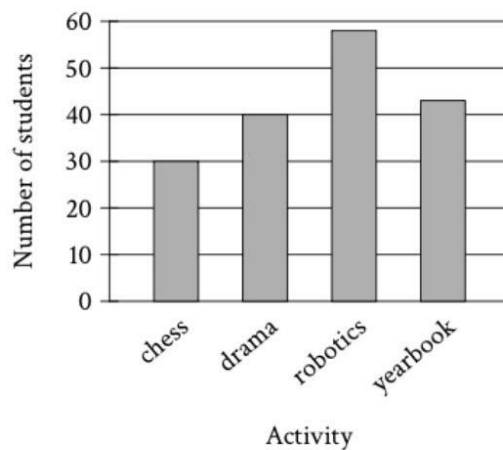
Question # ID
SAT9 M 2.4 a09ma204



The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is $(0, y)$. What is the value of y ?

SAT9 M 2.5 a09ma205

The bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school.



(A) 10

(B) 30

(C) 40

(D) 70

How many more students are in drama than in chess?

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Question # ID

SAT9 M 2.6 52931bfa

Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$?

A. $\frac{x-7}{5}$

B. $\frac{8x-3}{2}$

C. $\frac{8x^2-3x-14}{2x-14}$

D. $\frac{8x^2-3x-77}{2x-14}$

SAT9 M 2.7 eaab8bc1

Out of 300 seeds that were planted, 80% sprouted. How many of these seeds sprouted?

SAT9 M 2.8 68f2cbaf

Ty set a goal to walk at least 24 kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of 4 kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

A. 4

B. 6

C. 20

D. 24

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Question # **ID**
SAT9 M 2.9 83f2c3bf

$$y = x + 4$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	4
1	5
2	6

B.

x	y
0	6
1	5
2	4

C.

x	y
0	2
1	1
2	0

D.

x	y
0	0
1	1
2	2

SAT9 M 2.10 0b332f00

The function g is defined by $g(x) = 6x$. For what value of x is $g(x) = 54$?

SAT9 M 2.11 b8cbe394

Sean rents a tent at a cost of \$11 per day plus a onetime insurance fee of \$10. Which equation represents the total cost c , in dollars, to rent the tent with insurance for d days?

A. $c = 11(d + 10)$

B. $c = 10(d + 11)$

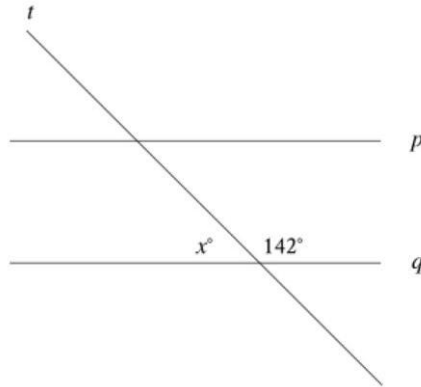
C. $c = 11d + 10$

D. $c = 10d + 11$

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Question # ID
SAT9 M 2.12 a09ma212



Note: Figure not drawn to scale.

In the figure, line p is parallel to line q , and line t intersects both lines. What is the value of $x + 142$?

- | | |
|--------|---------|
| (A) 52 | (C) 142 |
| (B) 90 | (D) 180 |

SAT9 M 2.13 575fle12

What is the area, in square centimeters, of a rectangle with a length of 34 centimeters (cm) and a width of 29 cm?

SAT9 M 2.14 a09ma214

$$h(x) = x + b$$

For the linear function h , b is a constant and $h(0) = 45$. What is the value of b ?

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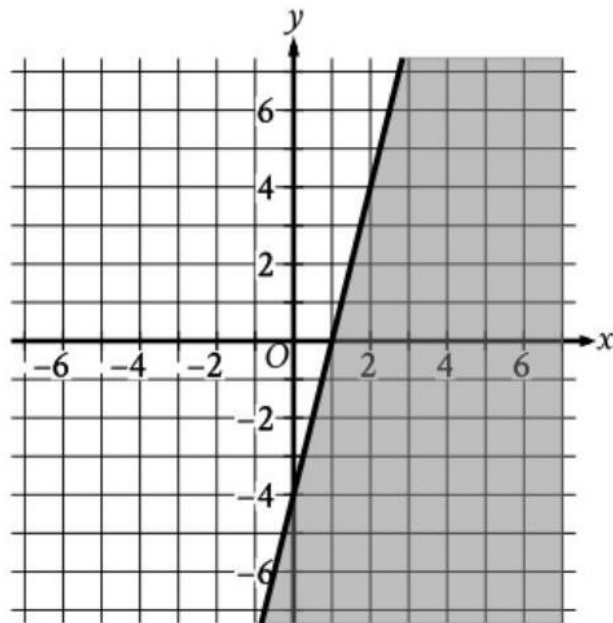
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Question # ID
SAT9 M 2.15 24854644

What is the equation of the line that passes through the point $(0, 5)$ and is parallel to the graph of $y = 7x + 4$ in the xy -plane?

- A. $y = 5x$
- B. $y = 7x + 5$
- C. $y = 7x$
- D. $y = 5x + 7$

SAT9 M 2.16 a09ma216



The shaded region shown represents the solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

(A) $(-5, -6)$

(C) $(1, 4)$

(B) $(-2, 5)$

(D) $(6, -2)$

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Question # ID

SAT9 M 2.17 7ba694f3

The number of bacteria in a liquid medium doubles every day. There are 44,000 bacteria in the liquid medium at the start of an observation. Which represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

A. $y = \frac{1}{2} (44,000)^t$

B. $y = 2(44,000)^t$

C. $y = 44,000\left(\frac{1}{2}\right)^t$

D. $y = 44,000(2)^t$

SAT9 M 2.18 a09ma218

The product of a positive number x and the number that is 8 more than x is 180. What is the value of x ?

Ⓐ 5

Ⓑ 10

Ⓒ 18

Ⓓ 36

SAT9 M 2.19 a09ma219

$$(5x + 4)(2x - 5) = 0$$

Which of the following is a solution to the given equation?

Ⓐ $-\frac{5}{2}$

Ⓑ $-\frac{5}{4}$

Ⓒ $-\frac{4}{5}$

Ⓓ $-\frac{2}{5}$

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Question # **ID**
SAT9 M 2.20 cc7ffe02

Keenan made 32 cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

SAT9 M 2.21 4993b828

The area A , in square centimeters, of a rectangular cutting board can be represented by the expression $w(w + 9)$, where w is the width, in centimeters, of the cutting board. Which expression represents the length, in centimeters, of the cutting board?

- A. $w(w + 9)$
- B. w
- C. 9
- D. $(w + 9)$

SAT9 M 2.22 fic1e971

The measure of angle R is $\frac{2\pi}{3}$ radians. The measure of angle T is $\frac{5\pi}{12}$ radians greater than the measure of angle R . What is the measure of angle T , in degrees?

- A. 75
- B. 120
- C. 195
- D. 390